

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Seventh Semester of B. Tech (EE) Examination****March-April 2018****EE403 Power Electronics & Drives-II****Date: 30/03/2018****Time: 10:00 a.m. To 01:00 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I**Q - 1 Answer the questions below. [07]**

a. An Inverter converts [01]

- (i) ac to dc (ii) dc to ac (iii) dc to dc (iv) ac to ac

b. In SPWM the modulating signal is [01]

- (i) Square (ii) Saw-tooth (iii) Sinusoidal (iv) Triangle

c. The ripple frequency of the output DC voltage of a 1- Φ rectifier supplied from a 230V, 50 Hz source is _____. [01]

- (i) 50 Hz (ii) 100Hz (iii) 150Hz (iv) 200Hz

d. Which of the following SMPS is used for low power application only? [01]

- (i) Push-Pull (ii) Fly-back (iii) Half Bridge (iv) Full Bridge

e. In an 12-pulse rectifier the dominant harmonics are _____. [01]

- (i) 7th & 9th (ii) 9th & 11th (iii) 11th & 13th (iv) 17th & 19th

f. State two applications of UPS. [02]

Q – 2.a What are Cycloconverters? Give its classification and explain its applications. [04]**Q – 2.b Answer any two questions. [10]**

(i) Draw and explain the working of a single phase full bridge square wave inverter.

(ii) What is Pulse width modulation technique? Draw and explain sine pulse width modulation technique, clearly describing how the output voltage and frequency can be varied by using it.

(iii) Explain 6-pulse and 12-pulse Diode based converter with circuit diagram and waveforms.

Q - 3 Answer any two questions. [14]a. Explain the basic principle of operation of a 1- Φ to 1- Φ step down midpoint Cycloconverter for continuous and discontinuous load.

- b. Explain 180° mode of conduction for a three phase VSI along with its necessary waveforms. Which mode of conduction is better 180° or 120° and why?
- c. List out the main functions of the phase-shifting transformer. Explain Y/Z-I phase-shifting transformer with necessary equations and vector diagram.

SECTION – II

Q - 4 Choose the correct answer. [07]

- a. AC voltage controllers convert [01]
 - (i) fixed ac to fixed dc (ii) variable ac to variable dc
 - (iii) fixed ac to variable ac (iv) variable ac to fixed ac
- b. The AC voltage controllers are used in _____ applications. [01]
 - (i) power generation (ii) electric heating
 - (iii) conveyor belt motion (iv) power transmission
- c. If k is the duty cycles of the controller, then the rms value of the output voltage in case of an integral cycle control circuit will be? Consider the input to be sinusoidal with peak value V_m and rms value V_s . [01]
 - (i) $V_s \times k$ (ii) V_s/k (iii) $V_s \times \sqrt{k}$ (iv) V_s
- d. Static UPS requires _____ [01]
 - (i) only rectifier (ii) only inverter (iii) both inverter and rectifier
 - (iv) none of the mentioned
- e. Applications of Cycloconverters include [01]
 - (i) speed control of ac drives (ii) induction heating
 - (iii) static VAR compensation (iv) all of the mentioned
- f. The single phase bridge type Cycloconverter uses _____ number of SCRs. [01]
 - (i) 4 (ii) 8 (iii) 6 (iv) none of the mentioned
- g. _____ is used for critical loads where temporary power failure can cause a great deal of inconvenience. [01]
 - (i) SMPS (ii) UPS (iii) MPS (iv) RCCB

Q – 5.a A single-phase ac voltage controller has input voltage of 230 V, 50 Hz and a load of $R=15\ \Omega$. For 6 cycles on and 4 cycles off, determine (i). RMS output voltage (ii). Input p.f (iii). Average and RMS thyristor currents. [06]

OR

Q – 5.a Draw and explain stator voltage control for three-phase induction motor with its characteristics. [06]

Q – 5.b Explain the advantages of multipulse converter. **[03]**

Q – 5.c List the various types of SMPS and explain any one configuration in detail. **[05]**

OR

Q – 5.c Draw and Explain principle of integral cycle control and find out equation of output rms voltage. **[05]**

Q – 6.a Describe the voltage control methods in a single phase inverter using external control of dc input voltage. **[04]**

Q – 6.b Draw the equivalent circuit of Induction motor and derive the expression for torque. **[06]**

Q – 6.c Describe single phase AC switches and DC switches in detail. **[04]**

OR

Q – 6.a What is an UPS? Describe online UPS and offline UPS configurations in detail. **[06]**

Q – 6.b Explain the working of a bridge-type step up Cycloconverter with circuit diagrams and waveforms. **[04]**

Q – 6.c A single-phase full wave ac voltage controller feeds a load of $R = 20\Omega$ with an input voltage of 230 V, 50 Hz. Firing angle for both the thyristors is 45° . Calculate (i) rms value of output voltage (ii) load power (iii) average and rms current of thyristors. **[04]**
